

BÁO CÁO THỰC HÀNH LAB 3

LẬP TRÌNH HƯỚNG ĐỐI TƯỢNG

Sinh Viên: Trần Mạnh Hùng – 20220068

Contents

1. Branch your repository.....	1
2. Working with Method Overloading	1
3. Passing parameter	3
4. Use debug run	4
5. Classifier Member and Instance Member	4
6. Open the Cart class	4
7. Implement the Store class.....	6
8. Re-organize your project.....	8
9. String, StringBuilder and StringBuffer	8
10. Release flow demonstration.....	11
11. Update UML Diagram	12

1.Branch your repository

```
PS C:\Users\84859\OneDrive\Documents\workspace\Java> git branch
main
* release/lab03
PS C:\Users\84859\OneDrive\Documents\workspace\Java>
```

2.Working with Method Overloading

2.1 Overloading by differing types of parameters

```

public void addDigitalVideoDisc(DigitalVideoDisc disc){
    if(qtyOrdered >= 20){
        System.out.println(x:"The cart is full");
    }
    else {
        itemsOrdered[qtyOrdered] = disc;
        qtyOrdered++;
        System.out.println(x:"The disc has been added");
        if(qtyOrdered >= MAX_NUMBERS_ORDERED - 2){
            System.out.println(x:"The cart is almost full");
        }
    }
}

```

- A new method that has the same name but with array type parameter

```

public void addDigitalVideoDisc(DigitalVideoDisc[] dvdList) {
    for (DigitalVideoDisc dvd : dvdList) {
        this.addDigitalVideoDisc(dvd);
    }
}

```

- A new method which allows to pass an arbitrary number of DVDs

```

public void addDigitalVideoDisc(DigitalVideoDisc... dvds) {
    for (DigitalVideoDisc dvd : dvds) {
        this.addDigitalVideoDisc(dvd);
    }
}

```

2.2 Overloading by differing the number of parameters

```
public void addDigitalVideoDisc(DigitalVideoDisc dvd1,DigitalVideoDisc dvd2) {  
    if (qtyOrdered + 2 <= MAX_NUMBERS_ORDERED){  
        itemsOrdered[qtyOrdered] = dvd1;  
        qtyOrdered++;  
        itemsOrdered[qtyOrdered] = dvd2;  
        System.out.println(dvd1.getTitle() + " and " + dvd2.getTitle() + " has been added!");  
        qtyOrdered++;  
    } else System.out.println(x:"Cannot add more DVDs! The cart is almost full.");  
}
```

3. Passing parameter

```
1 package hust.soict.dsai.test.disc;  
2  
3 import hust.soict.dsai.aims.disc.DigitalVideoDisc;  
4  
5 public class TestPassingParameter {  
6     Run | Debug  
7     public static void main(String[] args) {  
8         DigitalVideoDisc jungleDVD = new DigitalVideoDisc(title:"Jungle");  
9         DigitalVideoDisc cinderellaDVD = new DigitalVideoDisc(title:"Cinderella");  
10  
11         swap(jungleDVD, cinderellaDVD);  
12  
13         DigitalVideoDisc[] dvds = {jungleDVD, cinderellaDVD};  
14         swap(dvds);  
15  
16         System.out.println("jungle dvd title: " + jungleDVD.getTitle());  
17         System.out.println("cinderella dvd title: " + cinderellaDVD.getTitle());  
18  
19         System.out.println("jungle dvd title: " + dvds[0].getTitle());  
20         System.out.println("cinderella dvd title: " + dvds[1].getTitle());  
21  
22         changeTitle(jungleDVD, cinderellaDVD.getTitle());  
23         System.out.println("jungle dvd title: " + jungleDVD.getTitle());  
24     }  
25  
26     public static void swap(Object o1, Object o2){  
27         Object tmp = o1;  
28         o1 = o2;  
29         o2 = tmp;  
30     }  
31  
32     public static void swap(DigitalVideoDisc[] dvds) {  
33         DigitalVideoDisc tmp = dvds[0];  
34         dvds[0] = dvds[1];  
35         dvds[1] = tmp;  
36     }  
37  
38     public static void changeTitle(DigitalVideoDisc dvd, String title){  
39         String oldTitle = dvd.getTitle();  
40         dvd.setTitle(title);  
41         dvd = new DigitalVideoDisc(oldTitle);  
42     }  
43  
44
```

- A swap method that can correctly swap the two objects

```
public static void swap(DigitalVideoDisc o1, DigitalVideoDisc o2){  
    String tmp = o1.getTitle();  
    o1.setTitle(o2.getTitle());  
    o2.setTitle(tmp);  
}
```

4. Use debug run

5. Classifier Member and Instance Member

```
package hust.soict.dsai.aims.disc;  
  
public class DigitalVideoDisc {  
    private String title;  
    private String category;  
    private String director;  
    private int length;  
    private float cost;  
    private int id;  
  
    //static  
    private static int nbDigitalVideoDiscs = 0;  
  
    //getter  
    public String getTitle() {
```

6. Open the Cart class

- Create a new method printCart() to show cart info in given format

```

public void print(){
    System.out.println(x:"*****CART*****");
    System.out.println(x:"Ordered Items:");

    float totalCost = 0;

    for(int i = 0;i < qtyOrdered;i++){
        System.out.println((i + 1) + ". " + itemsOrdered[i].toString());
    }
    System.out.printf(format:"Total cost: %.2f $%n", totalCost());
    System.out.println(x:"*****");
}

```

- Search methods and isMatch method

```

public void searchById(int id){
    for(int i = 0;i < qtyOrdered;i++){
        if(itemsOrdered[i].getId() == id){
            System.out.println(itemsOrdered[i].toString());
            return;
        }
    }
    System.out.println("Not found DVD with Id = " + id);
}

public void searchByTitle(String title){
    for(int i = 0;i < qtyOrdered;i++){
        if(itemsOrdered[i].isMatch(title)){
            System.out.println(itemsOrdered[i].toString());
            return;
        }
    }
    System.out.println("Not found DVD with title : " + title);
}
}

```

```
    public boolean isMatch(String title){  
        return this.title.equalsIgnoreCase(title);  
    }  
}
```

7.Implement the Store class

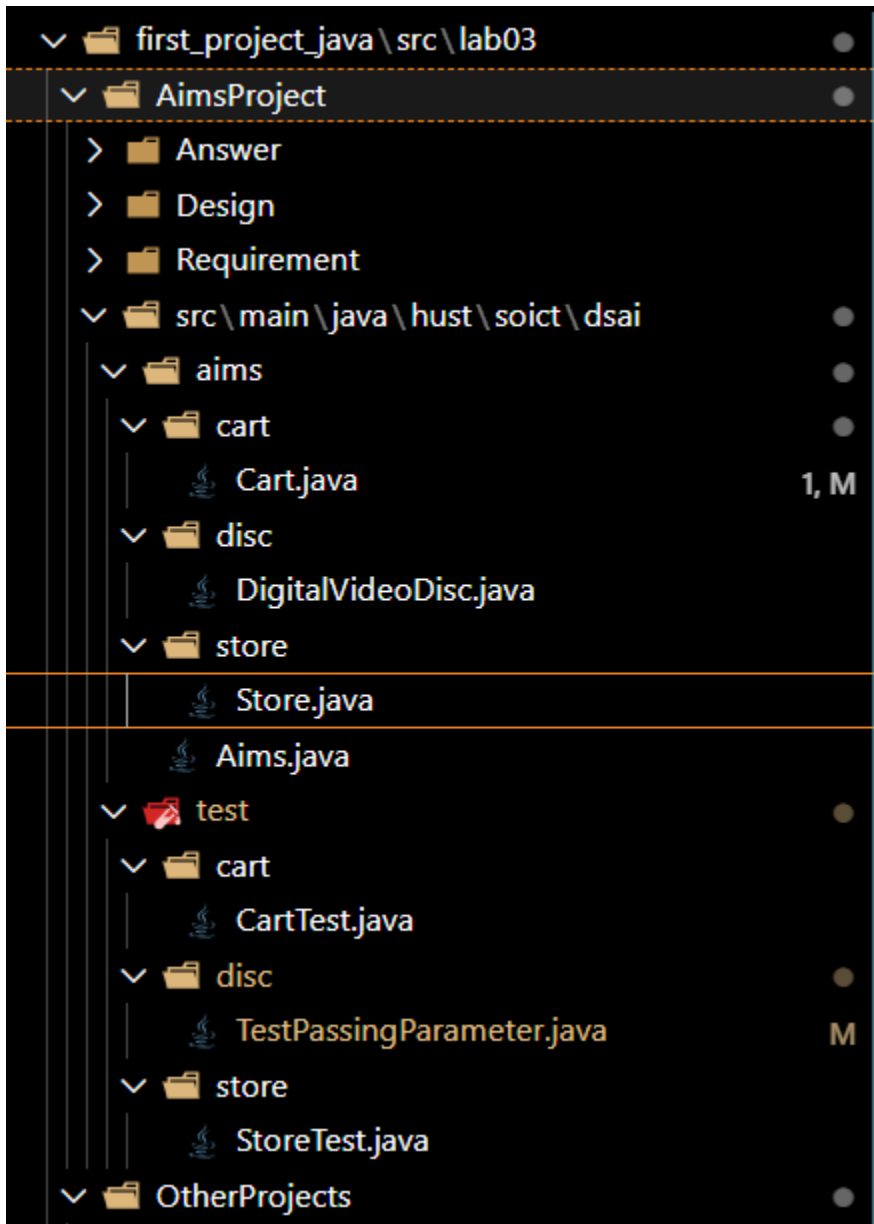
```

1 package hust.soict.dsai.aims.store;
2
3 import hust.soict.dsai.aims.disc.DigitalVideoDisc;
4
5 public class Store {
6     private static final int MAX_ITEMS_IN_STORE = 25;
7     private final DigitalVideoDisc[] itemsInStore = new DigitalVideoDisc[MAX_ITEMS_IN_STORE];
8     private int qtyInStore = 0;
9
10    public void addDVD(DigitalVideoDisc dvd){
11        if(qtyInStore >= MAX_ITEMS_IN_STORE){
12            System.out.println(x:"The store is full.");
13        }
14        else {
15            itemsInStore[qtyInStore] = dvd;
16            qtyInStore++;
17            System.out.println(x:"add DVD successfully");
18        }
19    }
20
21    public void removeDVD(DigitalVideoDisc dvd){
22        for(int i = 0; i < qtyInStore; i++){
23            if(itemsInStore[i] == dvd){
24                for(int j = i; j < qtyInStore - 1; j++){
25                    itemsInStore[j] = itemsInStore[j + 1];
26                }
27                itemsInStore[qtyInStore - 1] = null;
28                qtyInStore--;
29                System.out.println(x:"Successfully remove the DVD");
30                return;
31            }
32        }
33        System.out.println(x:"DVD not found in the store");
34    }
35
36    public void display(){
37        System.out.println(x:"*****STORE*****");
38
39        if(qtyInStore == 0){
40            System.out.println(x:"The store is empty.");
41        }
42        else {
43            for(int i = 0; i < qtyInStore; i++){
44                System.out.println((i + 1) + ". " + itemsInStore[i].getTitle());
45            }
46            System.out.println(x:"*****");
47        }
48    }
49 }

```

8.Re-organize your project

Rename project, use packages and re-organize hands-on labs and exercises.



9.String, StringBuilder and StringBuffer

```
1 package hust.soict.dsai.garbage;
2
3 import java.util.Random;
4
5 public class ConcatenationInLoops {
6     Run | Debug
7     public static void main(String[] args) {
8         Random r = new Random(seed:123);
9         long start = System.currentTimeMillis();
10        String s = "";
11        for(int i = 0; i < 65536; i++){
12            s += r.nextInt(bound:2);
13        }
14        System.out.println(System.currentTimeMillis() - start);
15
16        r = new Random(seed:123);
17        start = System.currentTimeMillis();
18        StringBuilder sb = new StringBuilder();
19        for(int i = 0; i < 65536; i++){
20            sb.append(r.nextInt(bound:2));
21        }
22        s = sb.toString();
23        System.out.println(System.currentTimeMillis() - start);
24    }
25 }
```

hust-project\java7\src\lab05\Other Projects\src\main\java\hust\soict\dsai\garbage

```
1 package hust.soict.dsai.garbage;
2
3 import java.io.IOException;
4 import java.nio.file.Files;
5 import java.nio.file.Paths;
6
7 public class GarbageCreator {
8     Run | Debug
9     public static void main(String[] args) {
10         String fileName = "Test/test.txt";
11         byte[] inputBytes = null;
12         long startTime, endTime;
13
14         try {
15             inputBytes = Files.readAllBytes(Paths.get(fileName));
16         } catch (IOException e) {
17             throw new RuntimeException(e);
18         }
19
20         startTime = System.currentTimeMillis();
21         String outputStr = "";
22         for(byte b : inputBytes){
23             outputStr += (char) b;
24         }
25         endTime = System.currentTimeMillis();
26
27         System.out.println("time: " + (endTime - startTime));
28     }
29 }
```

```

1 package hust.soict.dsai.garbage;
2
3 import java.io.IOException;
4 import java.nio.file.Files;
5 import java.nio.file.Paths;
6
7 public class NoGarbage {
8     Run | Debug
9     public static void main(String[] args) {
10         String fileName = "Test/test.txt";
11         long startTime, endTime;
12
13         byte[] inputBytes;
14         try {
15             inputBytes = Files.readAllBytes(Paths.get(fileName));
16         } catch (IOException e) {
17             throw new RuntimeException(e);
18         }
19         startTime = System.currentTimeMillis();
20         StringBuilder outputString = new StringBuilder();
21         for (byte b : inputBytes){
22             outputString.append((char) b);
23         }
24         endTime = System.currentTimeMillis();
25         System.out.println("time: " + (endTime - startTime));
26     }
27 }

```

10.Release flow demonstration

- Update local repository.
- Create, switch to a new branch in local repository and make modifications.
- Commit the change.

Class Diagram

